AIGR TreeGrafter Function-Inference Stress Test — Final Report

Target: *Heterocephalus glaber* (naked mole-rat) Cgas — UniProt **A0AAX6RS70** (554 aa) **Seed hypothesis (TreeGrafter/PANTHER, IEA, GO_REF:0000118):** Cgas has **2',3'-cyclic GMP-AMP synthase activity (GO:0061501) Verdict: SUPPORTED** — the propagated term is correct and already at maximal specificity. **RETAIN.**

Summary

The naked mole-rat protein A0AAX6RS70 ("Cgas") was electronically annotated with **2',3'-cyclic GMP-AMP synthase activity (GO:0061501)** by the TreeGrafter/PANTHER phylogenetic pipeline. The task was to independently judge whether this protein *directly* performs that function, and — because automated propagation can silently fail — to actively test three characteristic failure modes: (1) granularity (the term being a broad family function rather than this protein's specific subfamily activity), (2) pseudo-enzyme status (a conserved fold with a degenerate active site), and (3) within-superfamily mis-placement (grafting onto a structurally related but functionally distinct neighbor of the shared Mab-21 nucleotidyltransferase fold).

Across three iterations of analysis — domain-architecture inspection, global pairwise alignment to characterized human and mouse cGAS, residue-level active-site mapping, and a paralog-discrimination panel across the Mab-21 superfamily — **all three failure modes were explicitly tested and excluded.** The protein was grafted to the cGAS-specific subfamily node (PANTHER **PTHR10656:SF35, "CYCLIC GMP-AMP SYNTHASE"**), not the broad family node, so there is no granularity problem. Its complete catalytic apparatus — the three-acidic-residue Mg^{2+}-coordinating triad and the zinc thumb — is conserved with **zero substitutions, insertions, or deletions**, so it is not a pseudo-enzyme. And its nearest characterized neighbor, by a wide margin, is cGAS itself (≈57% global identity to both human and mouse cGAS), ruling out paralog mis-placement.

The conclusion is that the seed hypothesis is **SUPPORTED**: A0AAX6RS70 is a bona fide mammalian cGAS ortholog that directly performs the 2',3'-cGAMP synthase reaction (ATP + GTP → 2',3'-cGAMP). GO:0061501 should be **retained** at its current granularity; no generalization, sibling-replacement, or removal is warranted. The only residual limitation is the absence of a direct biochemical assay in *H. glaber* — all evidence is sequence/structure-based orthology inference, which is strong but formally inferential.

Key Findings

Finding 1 — A0AAX6RS70 is a genuine cGAS ortholog with fully intact catalytic machinery

The UniProt record for A0AAX6RS70 (554 aa, *Heterocephalus glaber*) carries the submission name "Cyclic GMP-AMP synthase isoform X1" and the canonical cGAS domain signature: Pfam **PF03281 (Mab-21)** plus **PF20266 (Mab-21_C)**, and InterPro **IPR046903** (Mab-21-like nucleotidyltransferase, residues 246–420) plus **IPR046906** (HhH / H2TH-like, residues 437–541). This is the two-lobed architecture of mammalian cGAS: an N-terminal nucleotidyltransferase (NTase) catalytic lobe fused to a C-terminal Mab-21 helix-hairpin-helix / zinc-thumb DNA-binding lobe.

Crucially, PANTHER placed the protein at **PTHR10656:SF35** = **"CYCLIC GMP-AMP SYNTHASE"** — a *subfamily-level* node specific to cGAS, not the broad pan-Mab-21 family node. The GO propagation therefore originated from the cGAS-specific clade, which immediately removes the most common granularity failure: the term inherited is the subfamily function, not a generic family function.

A Needleman–Wunsch global alignment (BLOSUM62) of A0AAX6RS70 against human CGAS (Q8N884) gives **57.2% whole-chain identity (297/519 aligned positions)** and **76.9% identity across the catalytic core** (human positions ~212–421). The whole-chain divergence is concentrated in the intrinsically disordered, low-complexity N-terminus (roughly residues 28–197), which is not part of the catalytic machinery and is expected to vary across species. Within the catalytic core, every functional residue is conserved:

Functional role	Human cGAS (Q8N884)	NMR Cgas (A0AAX6RS70)	Status
Mg^{2+}-coordinating catalytic acid	Glu225	**Glu257**	Conserved
Mg^{2+}-coordinating catalytic acid	Asp227	**Asp259**	Conserved
Mg^{2+}-coordinating catalytic acid	Asp319	**Asp351**	Conserved
Zinc thumb	His390	**His422**	Conserved
Zinc thumb	Cys396	**Cys428**	Conserved
Zinc thumb	Cys397	**Cys429**	Conserved
Zinc thumb	Cys404	**Cys436**	Conserved

There are **no substitutions, insertions, or deletions at any active-site, catalytic, or metal-binding position**, and the spacing between residues is preserved. The E–D–D triad coordinates the two divalent metal ions (Mg^{2+}, or Mn^{2+}) required for the two-metal-ion nucleotidyl-transfer chemistry, and the C_3H_1 zinc thumb stabilizes the activation loop and mediates DNA binding. A protein that conserves this entire apparatus in a mammal performs the mammal-specific 2',3'-cGAMP reaction. This excludes the pseudo-enzyme failure mode.

The canonical reaction is firmly established in the primary literature: cGAS is activated by cytosolic double-stranded DNA and uses ATP and GTP to synthesize the cyclic dinucleotide second messenger 2',3'-cGAMP (PMID: 42810574; PMID: 41806557). This reaction corresponds precisely to GO:0061501.

Finding 2 — The sequence is cGAS, not a Mab-21 family paralog (rules out within-superfamily mis-placement)

The Mab-21 superfamily contains several paralogs (MAB21L1, MAB21L2, MAB21L3, MB21D2) that share the NTase/Mab-21 fold but lack cGAS catalytic activity. To exclude the third failure mode — grafting onto a structurally related but functionally distinct neighbor — we computed the global Needleman–Wunsch percent identity of A0AAX6RS70 against each characterized superfamily member:

Query vs	% identity (aligned)	Note
cGAS (human, Q8N884)	**57.2% (297/519)**	nearest neighbor
cGAS (mouse, Q8C6L5)	**57.1% (287/503)**	ortholog, co-top hit
MAB21L3 (Q8N816)	45.7% (118/258)	short alignment; paralog
MAB21L2 (Q9Y586)	34.3% (123/359)	paralog, no cGAS activity
MAB21L1 (Q13394)	33.4% (120/359)	paralog, no cGAS activity
MB21D2 (Q8IYB1)	19.1% (94/491)	distant

The nearest neighbors are the two cGAS orthologs, and the margin over the next-closest paralog (MAB21L3) is large — ~12 percentage points, and the MAB21L3 match spans only 258 positions versus full-length coverage for cGAS. Over the next tier (MAB21L1/2) the margin is ~23 percentage points. Decisively, the **cGAS-diagnostic zinc thumb (His422/Cys428/Cys429/Cys436)** is present in A0AAX6RS70 — this motif is a hallmark of cGAS and is **absent** from the MAB21L1/2 paralogs that lack cGAS activity. The combination of (i) best global match to cGAS, (ii) full-length coverage of the catalytic core, and (iii) the cGAS-specific zinc thumb jointly excludes any within-superfamily mis-placement.

Mechanistic Model / Interpretation

The logical structure of the stress test — and how each failure mode was tested and excluded — is summarized below.

```
              TreeGrafter grafts A0AAX6RS70 → PANTHER PTHR10656:SF35
                         propagates GO:0061501 (IEA)
                                      |
          ------------------------------------------------------
          |                           |                        |
  (1) GRANULARITY?           (2) PSEUDO-ENZYME?         (3) MIS-PLACEMENT?
  Broad family node, so      Fold intact but active     Grafted onto a
  term is too general?       site degenerate?           different Mab-21
                                                        paralog, not cGAS?
          |                           |                        |
  Node = SF35, cGAS-         E257/D259/D351 triad       Best global match =
  SPECIFIC subfamily.        intact; zinc thumb         cGAS (57.2%), >12 pts
  No child term exists;      H422/C428/C429/C436        above nearest paralog;
  term already maximally     intact; 0 indels at        cGAS-specific zinc
  specific.                  active site.               thumb present.
          |                           |                        |
     NOT A PROBLEM               NOT A PROBLEM              NOT A PROBLEM
          ------------------------------------------------------
                                      |
                        VERDICT: SUPPORTED — RETAIN GO:0061501
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The domain architecture of cGAS (and of A0AAX6RS70):

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N-term (disordered, divergent)   NTase catalytic lobe        Mab-21 / zinc-thumb DNA-binding lo
|----------~245 aa------------|--IPR046903 (246–420)-----|--IPR046906 HhH/H2TH (437–541)--|
                                Mg2+ triad:                 Zinc thumb:
                                E257, D259, D351            H422, C428, C429, C436
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Biologically this is fully coherent. cGAS is the principal cytosolic dsDNA sensor of the cGAS–STING innate-immune axis. On binding dsDNA it undergoes allosteric activation and, using ATP and GTP with two catalytic metal ions, synthesizes 2',3'-cGAMP — the cyclic dinucleotide that binds STING and triggers type-I interferon signaling. The defining feature of the *mammalian* enzyme is the mixed 2'-5'/3'-5' phosphodiester linkage of its product (hence "2',3'-cGAMP"), which is exactly what GO:0061501 encodes. The naked mole-rat ortholog preserves every residue that executes this chemistry, and its sequence divergence is confined to the disordered, non-catalytic N-terminus, which is typical of cross-species cGAS comparison. Nothing in the sequence suggests a functional pivot to a different activity.

Evidence Base

#	Citation (PMID)	Evidence type	Role	Claim tested	Key finding	Context
1	Our analysis (UniProt A0AAX6RS70; InterPro/ Pfam; PANTHER SF35)	Computational / database	Supports	Correct subfamily graft?	Name "Cyclic GMP-AMP synthase isoform X1"; PF03281+PF20266; IPR046903+IPR046906; grafted to cGAS-specific node PTHR10656:SF35	*H. glaber*, 554 aa
2	Our analysis (NW alignment to Q8N884)	Structural / evolutionary (computational)	Supports	Catalytic residues conserved and spaced?	57.2% whole-chain / 76.9% catalytic-core identity; Mg^{2+} triad E257/D259/D351 and zinc thumb H422/ C428/C429/C436 all identical, 0 indels at active site	vs human cGAS
3	Our analysis (paralog panel)	Evolutionary (computational)	Supports / competing-excluded	Is this really a different paralog?	Nearest neighbor is cGAS (57.2%/57.1%) vs MAB21L3 45.7% (short span), MAB21L1/2 ~33–34%, MB21D2 19.1%; cGAS-specific zinc thumb present	Mab-21 superfamily
4	PMID: 42810574	Review / direct-assay basis	Supports	Exact cGAS reaction	"cyclic GMP-AMP synthase (cGAS) catalyzes ATP and GTP to synthesize the cyclic dinucleotide 2',3'-cGAMP"	mammalian cGAS
5			Supports		"structural basis of cGAS activation by	

#	Citation (PMID)	Evidence type	Role	Claim tested	Key finding	Context
	PMID: 41806557	Review / structural		Canonical activity	dsDNA and its catalytic synthesis of 2'3'-cGAMP"	mammalian cGAS–STING
6	PMID: 28940468	Direct assay / kinetic	Qualifies	Product distribution	First kinetic constants for 2',3'-cGAMP formation; notes 2',3'-cGAMP may be a *minor* product vs linear nucleotides in vitro	human cGAS
7	PMID: 32814054	Mechanistic	Qualifies	Metal dependence	DNA binding allosterically activates cGAS to synthesize 2'3'-cGAMP using Mg^{2+} (and Mn^{2+})	cGAS mechanism

How the literature supports the call. PMID: 42810574 and PMID: 41806557 jointly establish that the canonical, defining activity of mammalian cGAS is dsDNA-activated synthesis of 2',3'-cGAMP from ATP and GTP — i.e., exactly GO:0061501. PMID: 32814054 confirms the two-metal-ion mechanism that the conserved E257/D259/D351 triad serves. PMID: 28940468 provides a careful qualifier: in vitro, the 2',3'-cGAMP product can be minor relative to linear di-nucleotides — but this concerns catalytic efficiency and product ratio, not the identity of the enzyme or the validity of assigning GO:0061501. None of the reviewed literature challenges the assignment.

Active-Site / Placement Analysis (Provenance)

Method: Global pairwise Needleman–Wunsch alignment (BLOSUM62) of A0AAX6RS70 against human cGAS (Q8N884) and against a panel of characterized Mab-21 superfamily paralogs (Q8C6L5, Q13394, Q9Y586, Q8N816, Q8IYB1). Active-site positions were mapped from human cGAS numbering to the naked mole-rat sequence via the alignment. Sequences retrieved from UniProt REST; alignment executed in-session.

Catalytic / metal-binding residue conservation — verdict: FULLY INTACT

Site	Human position	NMR position	Conserved?
Mg^{2+} triad	Glu225	Glu257	✔
Mg^{2+} triad	Asp227	Asp259	✔
Mg^{2+} triad	Asp319	Asp351	✔
Zinc thumb	His390	His422	✔
Zinc thumb	Cys396	Cys428	✔
Zinc thumb	Cys397	Cys429	✔
Zinc thumb	Cys404	Cys436	✔

Subfamily placement — verdict: cGAS is the nearest characterized neighbor (see Finding 2 table). Catalytic-core identity to human cGAS is 76.9%; whole-chain 57.2% (297/519), with divergence confined to the disordered N-terminus.

Explicit statement: The catalytic and metal-binding residues are **intact and correctly spaced**, and the protein's nearest characterized neighbor is **cGAS**. Both independent tests converge on an active cGAS enzyme. No pseudo-enzyme signature; no paralog mis-placement.

GO Curation Implications

Recommended curator action (lead requiring verification): RETAIN GO:0061501 (2',3'-cyclic GMP-AMP synthase activity).

- The term is correct and at the right granularity — **same** granularity as the seed; it is a leaf-level molecular-function term encoding the diagnostic mixed 2'-5'/3'-5' linkage, so no more-specific child exists ("make-more-specific" not applicable).
- Do **not** generalize to a parent nucleotidyltransferase term — that would discard well-supported specificity.
- Do **not** replace with a sibling term — there is no evidence of a diverged or sibling activity; the paralog panel excludes MAB21L-type functions.
- Do **not** remove — the active site is intact and the ortholog assignment is unambiguous.
- Treat as a **core** molecular-function annotation for cGAS.

Because the current evidence code is **IEA** (electronic, GO_REF:0000118), the annotation remains inference-based. The convergent active-site conservation and paralog-distance evidence substantially raise confidence and could justify a curator-reviewed phylogenetic evidence note; a direct assay would be required to upgrade to an experimental code (IDA). "Protein binding" is explicitly *not* an appropriate fallback.

Limitations and Knowledge Gaps

1. **No direct biochemical assay in *H. glaber*.** All conclusions rest on orthology plus active-site conservation. Conserved residues strongly predict, but do not prove, catalytic competence and the exact 2',3' product regiochemistry.
2. **Alignment-based residue mapping.** Active-site positions were transferred from human cGAS numbering via global alignment rather than from an experimental naked mole-rat structure. An AlphaFold model superposition would add geometric confirmation but was not run.
3. **Product-ratio nuance.** Human-cGAS kinetics (PMID: 28940468) note that 2',3'-cGAMP can be a minor product relative to linear nucleotides in vitro. This concerns output proportions, not enzyme identity or the validity of GO:0061501.

4. **Database carry-over.** UniProt submission name and PANTHER/InterPro calls are themselves automated; however they are mutually consistent and corroborated by the independent active-site alignment, so paralog confusion (Mab21L1/2 or cGLR) is ruled out by the cGAS-diagnostic residue set.
5. **Species-specific innate-immune biology.** Naked mole-rats have notable immune and longevity peculiarities; whether NMR cGAS has altered regulation (DNA affinity, STING coupling) is unknown — relevant to downstream biological-process annotation but outside the scope of this molecular-function term.
6. **Isoform scope.** The record is "isoform X1"; alternative isoforms were not individually assessed.

Proposed Follow-up Experiments / Actions

1. **Direct activity assay (definitive).** Express recombinant NMR Cgas, activate with dsDNA, incubate with ATP/GTP + Mg^{2+}, and detect 2',3'-cGAMP by LC-MS/MS (with linkage analysis to confirm the 2'-5'/3'-5' bond) or via a STING-reporter activation readout. This is the single most efficient experiment to convert IEA → IDA and settle any residual doubt.
2. **AlphaFold structural superposition.** Superpose the AlphaFold model of A0AAX6RS70 on a human cGAS–DNA–nucleotide complex to confirm the 3D spacing of E257/D259/D351 and the zinc thumb, complementing the sequence-level conservation.
3. **Phylogenetic tree reconstruction.** Build a maximum-likelihood tree across the Mab-21 superfamily including the query, to visually confirm it falls inside the cGAS clade rather than at an internal paralog-splitting node — independent corroboration of the PANTHER SF35 graft.
4. **Curator note.** Record the active-site conservation table and paralog-distance panel as provenance supporting retention of GO:0061501, documenting that all three TreeGrafter failure modes were explicitly tested and excluded.

Conclusion

The TreeGrafter/PANTHER propagation of **GO:0061501** to naked mole-rat **Cgas (A0AAX6RS70)** is **correct**. The protein is a genuine mammalian cGAS ortholog grafted to the cGAS-specific subfamily node (PTHR10656:SF35), with the complete catalytic Mg^{2+}-coordinating triad (E257/D259/D351 = human E225/D227/D319) and zinc thumb (H422/C428/C429/C436 = human H390/C396/C397/C404) perfectly conserved, and with cGAS as its unambiguous nearest characterized neighbor (~57% global identity, ~12–23 points above the next paralog). None of the three characteristic failure modes — granularity, pseudo-enzyme, or within-superfamily misplacement — applies. **Curation action: RETAIN at current granularity.**

Provenance: UniProt REST (A0AAX6RS70, Q8N884, Q8C6L5, Q13394, Q9Y586, Q8N816, Q8IYB1); custom Needleman–Wunsch alignment (BLOSUM62) executed in-session; PubMed PMIDs 41806557, 42810574, 28940468, 32814054, 41738955.
